

Enhancing multivariate time series forecasting for container reloading operations: an improved adaptive multi-granularity cascade forest model integrating

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ABSTRACT

Despite advancements in multivariate time series forecasting, critical challenges persist in operational applications like railway port container reloading. Key limitations include inadequate raw data utilization and, fundamentally, an inability to model the intrinsic multi-granularity temporal patterns of logistics operations, which span from equipment cycles to weekly planning rhythms. To address these issues, this study proposes a novel Improved Multivariate Adaptive Multi-Granularity Cascade Forest Model (IM_MAGCFM). Its design is theoretically grounded to bridge specific gaps: (1) multi-granularity scanning aligned with operational cycles for meaningful time series decomposition; (2) repurposing Gradient Boosting Decision Trees as a feature derivation engine to capture complex interactions among factors like cargo type and weather; and (3) integrating XGBoost for feature selection to optimize the high-dimensional feature space, ensuring robustness and efficiency. Empirical analysis with Alashankou railway port data shows IM_MAGCFM significantly outperforms baselines, with average reductions of 57.83% in MSE, 36.08% in RMSE, 30.94% in MAE, and 67.86% in MAPE. Validation on an independent Malaszewicze port dataset confirms its generalizability and robustness across operational environments. This research contributes a domain-aware, theoretically coherent framework to multivariate forecasting, demonstrating how logistics insights can drive integrated machine learning design for enhanced predictive accuracy and reliable operational decision-making in international rail logistics.

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1. Introduction

1.1. Background and motivation

The China Railway Express (CRE) emerges not merely as a logistical conduit but as a vital artery pulsating at the heart of the intricate supply chain that interlinks China with the vast expanse of Europe. As the ambitious tapestry of the “One Belt, One Road” initiative unfolds, the CRE's role morphs into an increasingly crucial linchpin, one that is both strategic and economically significant (Wei & Lee, 2021). In the realm of railway transport, the narrative is complicated by the stark contrast in railway gauges: China's network, meticulously engineered to a standard gauge of 1435 mm, stands in juxtaposition to the broad-gauge of 1520 mm that dominates Central Asia and Mongolia. This disparity is not just a technical detail; it is a formidable barrier that necessitates the intricate ballet of container reloading operations at transshipment yards, particularly at key railway ports. Fig. 1 illustrates the container reloading operations at the Alashankou Pass border station for the CRE route from Chengdu to Europe. Here, the meticulous transfer of containers to freight trains that conform to the destination country's railway specifications becomes a critical operation. This process is not merely procedural; it is the linchpin ensuring the seamless transportation of goods across the diverse and often disparate tracks of multiple nations (Bortfeldt & Wäscher, 2013). In practice, the duration of these container reloading operations is affected by several factors, including the available facilities and equipment at the railway ports, the proficiency of the staff, the productivity of container handling, and the efficiency of the customs clearance system.

The precise prediction of container reloading time goes beyond its technical definition; it is a vital factor in enhancing port operational efficiency and multimodal supply chain performance. Accurate forecasts are crucial for several key managerial functions. Firstly, they allow for dynamic resource allocation, optimizing equipment deployment and labor scheduling in accordance with anticipated operational demand, which helps minimize resource idleness and alleviate congestion. Secondly, they support integrated schedule planning, ensuring the synchronization of inbound train arrivals, yard operations, and outbound departures—essential for maintaining corridor schedule integrity and reducing asset detention. Lastly, they improve supply chain visibility and reliability, decreasing uncertainty for logistics stakeholders and bolstering the competitive value proposition as a competitive international freight corridor (Archetti et al., 2022). As the China–Europe Railway Express expands under the Belt and Road Initiative, developing innovative forecasting methods and technological solutions becomes increasingly important. Addressing these challenges will help optimize China–Europe logistics, improve efficiency, and strengthen the reliability of international trade.

1.2. Multimethodological approach and gaps

Historically, researchers have leaned on traditional statistical techniques to model container reloading times. Regression analysis has often served as the backbone of these efforts. For instance, multiple linear regression has been applied to assess factors such as equipment efficiency and scheduling (Chu et al., 2024). However, such models often fall short in capturing real-world complexity due to their linear assumptions.

The emergence of machine learning (ML) has revolutionized time prediction methodologies. Researchers have harnessed various algorithms, from regression models to neural networks, to analyze historical data. Algorithms such as Random Forests (RFs) and Gradient Boosting Machines (GBMs) have gained traction for their ability to model nonlinear relationships among multiple variables. A pivotal study demonstrated that random forests significantly outperformed traditional regression models in predicting container reloading times (Abreu et al., 2023), effectively handling interactions and nonlinearities within the data.

The advent of deep learning has further transformed the landscape of time prediction. Recurrent Neural Networks (RNNs) and their advanced variant, Long Short-Term Memory (LSTM) networks, have become indispensable tools for capturing temporal dependencies in sequential data. Research by Yan et al. (2024) illustrated the effectiveness of LSTMs in forecasting reloading times, leveraging vast datasets to identify patterns those traditional methods might overlook. Moreover, the integration of Convolutional Neural Networks



Fig. 1. The container reloading operations at the Alashankou Pass border station. Retrieved from: http://www.china-railway.com.cn/gjhz/zoblys/201910/t20191031_97163.html.

(CNNs) for feature extraction has shown promise, enhancing the predictive power of hybrid models (Kumar et al., 2025).

Despite notable advancements, several challenges persist in accurately predicting container reloading times. Data quality and availability remain significant obstacles, as historical datasets are often riddled with inconsistencies, missing values, and outliers, which can compromise model training and lead to unreliable predictions (Guo et al., 2025a,2025b). Furthermore, the curse of dimensionality complicates the modeling of high-dimensional datasets, where relationships among influencing factors—such as weather, equipment status, and personnel efficiency—become increasingly intricate (Thudumu et al., 2020). Compounding the myriad data-related challenges faced in railway port operations is a fundamental, domain-specific characteristic: the inherent multi-granularity temporal patterns intricately tied to the rhythms of train scheduling. The reloading process, far from being a monolithic entity, unfolds in a hierarchical manner, with its efficiency and duration dictated by dynamics operating at various scales (Guo et al., 2026).

- (1) Micro Granularity: At this level, we focus on individual train runs, where the unique composition and real-time conditions of an arriving train dictate operational outcomes.
- (2) Meso Granularity: Here, we consider the weekly cycle, shaped by consolidated schedules, resource allocation strategies, and recurring weekly patterns that influence operational flow.
- (3) Macro Granularity: This broader perspective encompasses bi-weekly or phase cycles, where longer-term cargo flow trends, maintenance schedules, and seasonal adjustments come into play.

Most existing forecasting models, typically designed around a single, uniform temporal scale, fall short in capturing these interconnected, schedule-driven hierarchical patterns. This oversight creates a significant chasm between model capabilities and the phased, batch-oriented realities of port operations.

To navigate the complexities inherent in these operational rhythms, recent studies have increasingly embraced the integration of diverse, high-dimensional data sources—from historical operational records to real-time environmental feeds. By amalgamating information from sensors, weather forecasts, and traffic reports, researchers have developed more nuanced models. For instance, incorporating detailed weather patterns has been shown to significantly improve prediction accuracy, underscoring the value of rich datasets for operational efficiency (Filom et al., 2022). This evolving approach highlights a critical pathway forward, yet it also demands forecasting frameworks capable of leveraging such multi-source data while simultaneously resolving the core challenge of modeling multi-scale, schedule-based temporal dependencies. A summary of key forecasting models applied in container reloading operations, illustrating this methodological evolution, is provided in Table 1.

Despite the promising advancements in these studies, several critical deficiencies remain unsolved.

- Insufficient utilization of raw time-series information.

Many research methodologies fail to fully exploit the raw information contained in multivariate time series data. Raw time series data encapsulates valuable historical trends and dynamic changes essential for accurate time predictions. However, limitations in data processing techniques often lead to the neglect of this information. Furthermore, the failure to leverage raw data effectively can significantly hinder the accuracy and reliability of predictions, resulting in suboptimal decision-making in operational contexts.

- Loss of Important Hidden Attribute Information

Certain critical pieces of information within time series data may possess hidden attributes that are challenging to extract using

Table 1
Key forecasting models utilized in container reloading operations.

Types	Models	Non-stationarity	Long-term dependencies	Inter-sequence similarity	Parameter complexity	Problem-solving capabilities
Traditional statistics-based techniques	MA	×	×	×	Low	Linear
	ARMA (Wang et al., 2016)	×	×	×	Low	Linear
	ARIMA (Xie et al., 2022)	✓	✓	×	Medium	Linear
	Monte Carlo (El-Ouardi et al., 2021)	✓	×	×	Medium	Nonlinear
Deep Learning –based techniques	LSTM (Yang et al., 2019)	✓	✓	✓	Medium	Nonlinear
	CNN (Luo et al., 2019)	✓	×	✓	Medium to High	Nonlinear
	GBM (Laifa et al., 2021)	✓	✓	✓	Medium to High	Nonlinear
Combination-based techniques	Heuristic algorithm (Kim & Yi, 2021)	✓	✓	✓	High	Mixed linear and nonlinear
	SARIMA-SVR (Sadeghi Gargari et al., 2022)	✓	✓	✓	High	Mixed linear and nonlinear
	CNN-Bi_LSTM (Guo et al., 2022)	✓	✓	✓	High	Mixed linear and nonlinear

Note: MA represents moving average model; ARMA represents autoregressive moving average model; ARIMA represents autoregressive integrated moving average model; SARIMA-SVR represents Seasonal ARIMA with support vector regression.

conventional methods. Important information may be embedded in the deeper structure of the data or exist in complex relationships among multiple variables. This complexity can lead to the inadvertent loss of crucial insights during preprocessing and analysis. Moreover, the oversight of these hidden attributes can diminish the performance of predictive models, as they are unable to capture the full scope of the underlying dynamics.

- **Difficulty in Capturing Multi-Scale Variations**

Time series data often exhibit variations across multiple scales, presenting unique challenges for prediction models: (1) data can display distinct patterns and trends at various time scales, making it challenging for existing multivariate prediction methods to effectively capture these changes; (2) the inability to accurately reflect real dynamic changes in data with complex time scale characteristics can lead to subpar model performance, thereby affecting the accuracy and applicability of predictions. As previously discussed, these challenges are particularly pronounced in the realm of container reloading operations, where the intrinsic multi-granularity nature of the processes amplifies the complexities involved. The need for robust predictive models that can adeptly navigate these intricacies is critical for enhancing operational efficiency and decision-making.

1.3. Contribution and paper organization

In the above analysis, it is evident that multivariate forecasting requires consideration of the interactions and relationships among multiple variables. These variables may originate from different data sources and possess varying time scales and frequencies, making the exploration of their interconnections and mutual influences particularly complex. Furthermore, when predicting container reloading times, factors such as cargo type, reloading equipment status, operator skill levels, weather conditions, and traffic situations must be considered. The intricate interactions among these factors complicate the data association mining process. Moreover, multivariate time series forecasting models must handle multiple input variables and produce accurate predictions. This necessitates a high degree of flexibility and adaptability in the model to capture the dynamic relationships between variables. Importantly, the model must be explicitly designed to accommodate the multi-granularity temporal patterns intrinsic to operations, as identified in [Section 1.2](#). This presents a significant challenge in selecting appropriate algorithms and structural frameworks during model construction, ensuring that the model faithfully reflects the inherent patterns and trends present in the data. Additionally, the complexity and uncertainty that characterize multivariate time series data introduce substantial hurdles for model construction and optimization. Navigating these challenges is crucial for developing robust forecasting models that enhance operational efficiency and decision-making processes.

To address these issues, we propose an improved multivariate adaptive multi-granularity cascade forest model that integrates various techniques, including adaptive cascade forests, multi-factor feature selection, and multi-granularity scanning, aimed at enhancing prediction accuracy and robustness for container reloading operations at CRE. The main contributions are outlined as follows:

1. To explore the relationships among various factors and container reloading times, we have developed an improved multivariate adaptive multi-granularity cascade forest model (IM_MAGCFM) that incorporates adaptive cascade forests, multi-factor feature selection, and multi-granularity scanning techniques. This model is specifically designed to address the multi-source data structures associated with large railway ports' container reloading time prediction.
2. The Gradient Boosting Decision Tree (GBDT) algorithm is designed for feature derivation from the raw time series data, enriching the dataset's information representation capabilities and uncovering deeper, more subtle relationships among feature time series. This approach addresses the challenges of predicting container reloading times at large ports, which are often complicated by the difficulty of obtaining feature data and the presence of numerous uncertainty factors.
3. In terms of feature selection, we propose a gain-based XGBoost feature importance evaluation method as a multidimensional time feature selection technique for container reloading operations. This method assesses the contribution of each feature to the reloading time, allowing us to identify features that significantly impact model training and prediction. This module reduces the dimensionality of the feature space, enhances computational efficiency, and mitigates the risk of model overfitting, thereby improving the model's generalization capability.
4. We present an innovative multi-granularity scanning mechanism tailored to decompose time series data based on the hierarchical temporal patterns of railway operations. This method meets the need for integrated modeling across operational, planning, and strategic time scales. The resulting subsequences effectively capture the multi-scale temporal characteristics of container reloading times, facilitating multidimensional predictions that enhance forecast accuracy by accurately representing the phased nature of port operations.
5. To our knowledge, this is the first study focused on predicting container reloading times at large railway ports with multi-source data structures. Utilizing actual data from the Alashankou reloading operations, we validate the advanced nature and superiority of the proposed model.

The remainder of this paper is structured as follows: [Section 2](#) provides a concise literature review of multivariate time series forecasting methods, feature selection and feature derivation. [Section 3](#) outlines the framework and algorithm design of the IM_MAGCFM model. [Section 4](#) presents the experimental results obtained from the constructed model. Finally, [Section 5](#) summarizes the research findings and suggests directions for future research.

2. Literature review

2.1. Classical statistical forecasting methods

The early development of time series forecasting was predominantly grounded in classical statistical techniques. Models such as the Moving Average (MA), Autoregressive Moving Average (ARMA) (Anderson et al., 2016), and their more sophisticated extensions, including the Autoregressive Integrated Moving Average (ARIMA) (Xu et al., 2019) and Seasonal ARIMA (SARIMA) (Zhang et al., 2020), formed the cornerstone of this era. These methods, primarily based on the autocorrelation structure within a single time series, demonstrated strong efficacy in addressing univariate forecasting problems. However, their inherent linear assumptions and limited capacity to model complex, nonlinear interdependencies often led to suboptimal performance when confronted with the intricate, high-dimensional relationships characteristic of multivariate time series data (Kong et al., 2024).

2.2. Machine learning-based forecasting methods

The limitations of traditional statistical models catalyzed a significant shift towards machine learning (ML) methodologies. Algorithms such as Support Vector Machines (SVM), Random Forests (RF) (Guo et al., 2022), and Gradient Boosting Machines (GBM) gained prominence for their superior ability to capture nonlinear patterns and interactions among multiple variables. Ensemble learning techniques, exemplified by frameworks like the Hierarchical Vote Collective of Transformation-based Ensembles (HIVE-COTE) (Bagnall et al., 2015), further advanced the field by synergizing multiple learners to enhance prediction robustness. Despite these advancements, challenges persist, including model interpretability, computational complexity, and difficulties in effectively modeling long-term temporal dependencies and high-order feature interactions inherent in complex operational data.

2.3. Deep learning-based forecasting methods

The advent of deep learning has revolutionized time series forecasting, particularly through architectures designed for sequential data. Recurrent Neural Networks (RNNs) and their advanced variants, such as Long Short-Term Memory (LSTM) networks (Liu et al., 2023) and Gated Recurrent Units (GRU) (Chuwang et al., 2023), excel at capturing long-term dependencies. More recently, Transformer-based and transfer learning models have shown promise in modeling complex temporal patterns (Guo et al., 2024). While these models offer powerful representational capacity, they often require large volumes of data, extensive hyperparameter tuning, and substantial computational resources. Moreover, they can struggle with the joint modeling of multi-scale temporal variations and the interpretable integration of heterogeneous, high-dimensional features commonly found in logistics operations (Zhao et al., 2025).

2.4. Feature engineering for time series forecasting

Effective feature engineering is crucial for enhancing model performance, encompassing both feature selection and feature derivation.

2.4.1. Feature selection

Feature selection stands as a pivotal step in the realms of data mining, ML, and time series analysis. At its core, the principle of feature selection revolves around identifying the most valuable subset of features from an original feature set, thereby reducing model complexity and enhancing predictive performance. This intricate process is not merely a technical necessity; it is a strategic maneuver that can significantly influence the efficacy of predictive models (Zong et al., 2015). Common methods of feature selection include:

- **Filter Methods:** These rely on statistical metrics to assess the relationships between features and target variables, utilizing techniques such as correlation coefficients and mutual information (Li et al., 2017). While filter methods are computationally efficient, they often fall short in capturing complex multivariate interactions among features, and their results can be heavily influenced by the underlying data distribution (Walling & Lodh, 2024).
- **Wrapper and Embedded Methods:** These methods integrate feature selection into the model training process itself. Notable approaches include recursive feature elimination and model-based importance scoring, which employ algorithms like Random Forest or XGBoost to evaluate feature contributions during modeling (Demir & Şahin, 2022). By accounting for feature interactions, these methods typically yield more precise feature subsets. However, they incur significant computational overhead, and selected features often remain tied to specific models, limiting their generalizability (Xu et al., 2023).
- **Deep Learning-Based Methods:** These techniques automatically figure out which features are important using attention mechanisms or special designs in neural networks. While they offer powerful capabilities, they are often perceived as black boxes, sacrificing interpretability for performance. Additionally, they typically require large datasets and substantial computational resources (Bao et al., 2025).

To sum up, the art of feature selection transcends mere technical execution; it embodies a confluence of methodical rigor, domain knowledge, and strategic foresight. The challenge lies in overcoming the limitations of existing methods while integrating business insights, data exploration, model specificity, and appropriate evaluation metrics. This multifaceted approach to feature selection is not just a technical hurdle; it is a crucial technological endeavor that this paper seeks to address, aiming to refine the process and enhance

predictive capabilities across various applications.

2.4.2. Feature derivation

Feature derivation is a vital process that involves creating new, informative attributes from raw data to uncover underlying patterns (Sekar et al., 2025). This process encompasses a range of approaches, from traditional techniques to modern learning-based methods.

Traditional feature derivation typically includes generating statistical summaries, such as rolling means and variances, extracting seasonal components, and constructing lagged or differenced features (de Jesus et al., 2025). While these methods are intuitive and widely applicable, they have notable limitations. Noise and outliers can significantly affect them, and they often heavily rely on prior domain knowledge (Jiang et al., 2024). Additionally, they may struggle to capture complex, nonlinear relationships within the data (Wang et al., 2024a). The selection of critical parameters, such as lag order, often relies on subjective judgment, which can undermine the consistency and robustness of the derived features (Yang et al., 2023).

In contrast, model-based feature derivation leverages advanced machine learning algorithms to automatically learn higher-level feature representations. Gradient boosting decision trees generate effective nonlinear feature combinations through hierarchical splitting rules (Chakraborty, 2024), while autoencoders learn compressed, informative data encodings (Yang & Hu, 2024). These advanced approaches demonstrate a strong capability for uncovering sophisticated, latent patterns. However, they concomitantly introduce significant risks, including heightened model overfitting, increased computational complexity, and reduced interpretability. These challenges are particularly pronounced when dealing with limited or noisy datasets (Wang et al., 2024b).

Therefore, it is essential to address these shortcomings by selecting appropriate feature derivation methods tailored to specific datasets and business contexts. This represents a pressing technical challenge in the field of ML and data analysis.

2.5. Summary

The preceding analysis reveals persistent challenges in applying existing forecasting methodologies to intricate real-world problems, such as predicting container reloading times. This task is characterized by multivariate, multi-source, and multi-scale data, presenting several key issues:

Insufficient Nonlinear Relationship Capture: Many existing models still struggle to fully encapsulate the nonlinear dynamics present in complex datasets.

Challenges in High-Dimensional Feature Interaction Modeling: The intricacies of modeling interactions among high-dimensional features remain a formidable obstacle.

Inability to achieve joint modeling of multi-granularity temporal patterns: Current methodologies often fail to integrate various temporal resolutions effectively.

To address these interconnected gaps, this study proposes a novel integrated framework. The Deep Forest model (Zhou & Feng, 2019) serves as a foundational inspiration due to its hierarchical ensemble structure, robustness, and lower sensitivity to hyper-parameters. Building on this foundation, we present the Improved Multivariate Adaptive Multi-Granularity Cascade Forest Model (IM_MAGCFM), which strategically synthesizes several key innovations:

Adaptive Cascade Forest Structure: This component enhances the modeling of nonlinear relationships (Addressing Gap 1).

Hybrid Feature Engineering Pipeline: Utilizing Gradient Boosting Decision Trees (GBDT) for advanced feature derivation and

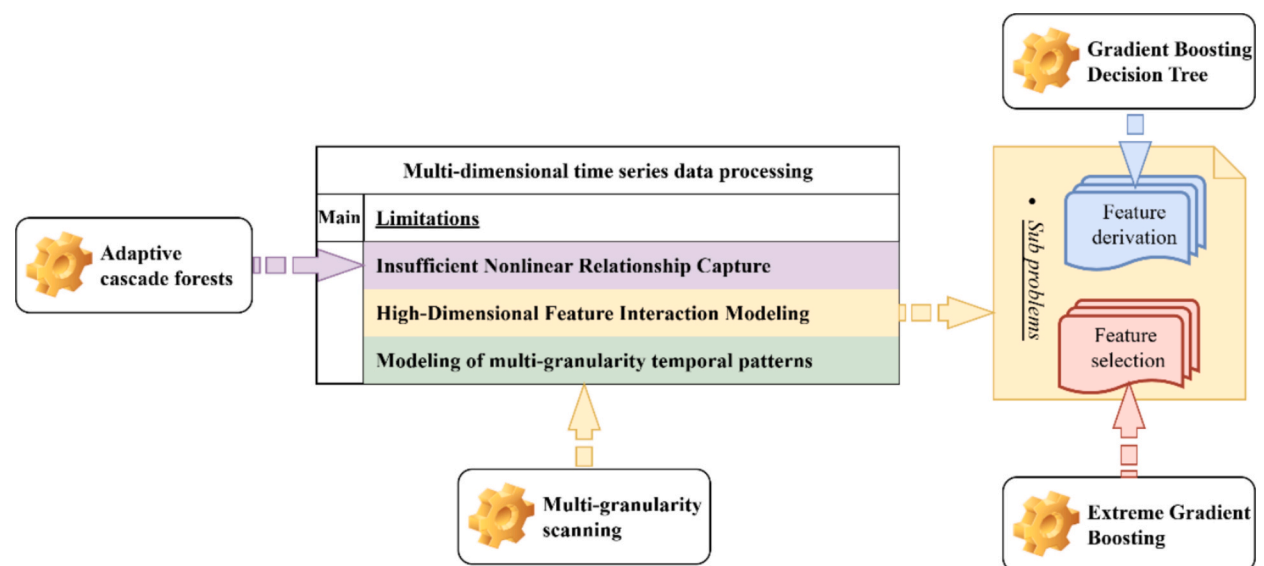


Fig. 2. Research framework based on proposed solutions.

XGBoost for gain-based feature selection, this pipeline effectively manages high-dimensional interactions (Addressing Gap 2).

Enhanced Multi-Granularity Scanning Mechanism: Designed to decompose and jointly analyze temporal patterns across different scales, this mechanism addresses the challenges of integrating multi-granularity temporal patterns (Addressing Gap 3).

This comprehensive approach aims to leverage the strengths of adaptive ensemble learning and targeted feature engineering, ultimately achieving superior forecasting accuracy and robustness in complex logistics operations. The overarching research framework that connects these solutions to the identified gaps is illustrated in Fig. 2.

3. Methodology

3.1. Multi-grained cascade forest

The multi-grained cascade forest (GcForest) algorithm provides a robust ensemble framework for time series analysis. However, its conventional formulation exhibits a critical limitation when applied to domain-specific forecasting problems: the selection of sliding window dimensions lacks principled grounding in the underlying temporal structure of the target domain. While the theoretical interpretation of multi-grained scanning as a multi-resolution analysis remains valid, this general formulation fails to address the specific temporal hierarchies inherent in operational contexts such as railway port logistics.

To bridge this methodological gap, the present study reconceptualizes the multi-grained scanning mechanism from a purely algorithmic tool to a domain-informed analytical process. In the application to container reloading time prediction, the sliding window dimensions are aligned with the fundamental operational cycles identified through domain analysis in Section 1.2. Specifically, window dimensions of $\{1, 7, 14\}$ time units are employed, which correspond directly to three critical operational scales: individual train processing cycles (micro-operational scale), weekly resource planning cycles (*meso*-planning scale), and bi-weekly operational phase transitions (macro-strategic scale). This domain-aligned configuration transforms the scanning mechanism from a generic feature extraction tool into a structured representation learning process that preserves the temporal semantics of port operations.

Fig. 3 illustrates the complete process of deep forest modeling, which encompasses both multi-grained scanning and a cascade forest structure. Suppose the dataset contains m samples, each with 400-dimensional features. The sliding window granularity is selected as 100, 200 and 300 dimensions to deal with the three-class classification problem. Scanning with a sliding window of width 100 dimensions results in 100-dimensional subsamples. Subsequently, all samples are fed into both the random forest and completely random forest for training, producing $301 \times 3 \times 2 = 1860$ -dimensional feature vectors. Similarly, a window size of 200 produces a 1206-dimensional feature vector, while a window size of 300 results in a 606-dimensional feature vector.

The feature vectors generated from the multi-granularity sliding window processing are then used as inputs for the first level of the cascade hierarchy. Assuming that each level of the cascade forest consists of four forest-based learners and is designed for a three-class classification problem, the output feature length from each level after training all forests is 12 dimensions. These 12 dimensions, when concatenated with the input dimensions, serve as input features for the subsequent level. This process continues until the test set indicates that further deepening of the cascade should cease. The output layer obtains the final prediction by averaging the class vectors produced at the last level of the cascade.

3.2. GBDT algorithm

While the multi-grained cascade forest provides a powerful framework for temporal pattern recognition, its standard implementation exhibits a fundamental representational limitation: it processes features sequentially through hierarchical layers without

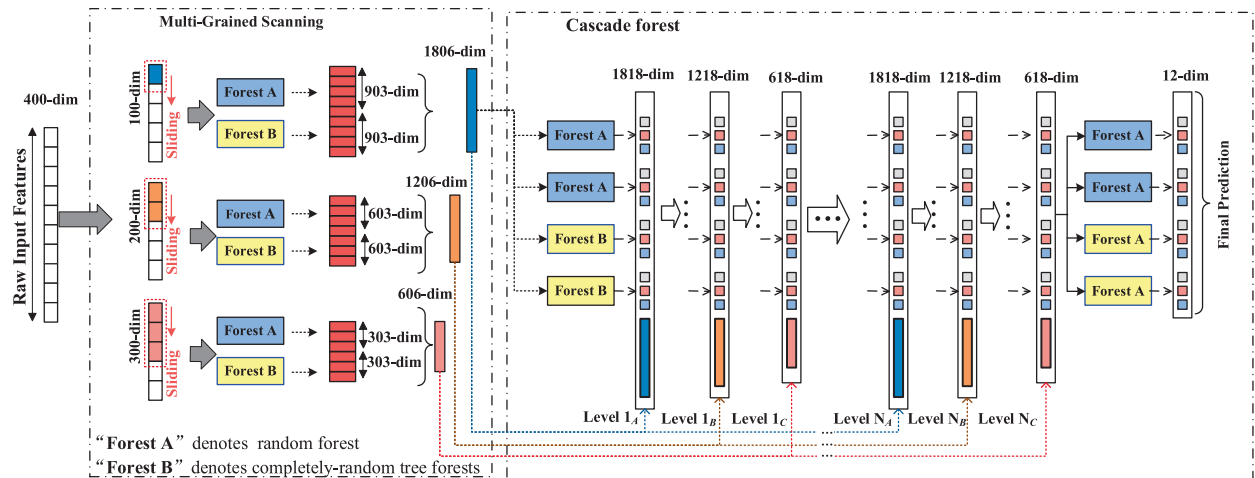


Fig. 3. GcForest frame diagram.

actively constructing new feature representations that capture complex interactions among operational variables. This limitation is particularly consequential in the present application context, where container reloading duration emerges from intricate, non-linear interactions among factors such as equipment availability, weather conditions, personnel allocation, and operational protocols.

To overcome this representational constraint, the present study introduces Gradient Boosting Decision Trees (GBDT) as a feature derivation mechanism that actively constructs higher-order feature representations through implicit feature crossing. Fig. 4 illustrates the GBDT algorithm flow; employing GBDT for feature derivation provides a principled approach to modeling complex, high-order interactions among operational factors. Each decision tree constructs a series of hierarchical splits based on conditions like $x_f < \theta$, effectively partitioning the feature space. A sample's path to a specific leaf node represents a unique combination of these split rules. By using the index of the terminal leaf across all trees as new binary features, the original input space is transformed into one where nonlinear interactions are explicitly encoded. This process performs implicit feature crossing, generating a more expressive representation that is crucial for capturing the intricate couplings between variables such as equipment status, weather conditions, and operator efficiency. Once the GBDT model is trained, it can transform the original feature vector into a new discrete feature vector. Fig. 5 illustrates the specific transformation process.

During GBDT training, each sample is assigned to a leaf node according to the decision rules for each node. The assigned leaf node is labeled as 1, while all other leaf nodes are labeled as 0. These leaf nodes collectively form the feature vector for that subtree. Subsequently, the feature vectors from all subtrees are concatenated to create the final feature vector, which serve as input for the subsequent model.

This paper employs the GBDT algorithm for the feature derivation process. By utilizing the GBDT algorithm, original features can be combined and transformed to generate new features. This approach enhances the model's ability to capture complex relationships and provides richer information for subsequent model inputs.

3.3. XGBoost algorithm

The GBDT-based feature derivation process effectively captures complex interactions but also presents a challenge: the one-hot encoding of leaf node indices leads to high-dimensional, sparse feature representations. This can complicate subsequent learning processes due to increased computational complexity and the risk of overfitting, highlighting a classic trade-off in representation learning between expressive capacity and efficiency.

To address this challenge while preserving the informative feature interactions captured by GBDT, the present study introduces XGBoost as a feature selection mechanism that operates on the derived feature space (Zhang et al., 2025). Unlike filter methods that evaluate features independently of the learning algorithm, or wrapper methods that are computationally intensive, XGBoost provides an embedded feature selection approach through its gain-based importance metric.

The theoretical innovation of this approach lies in its application of XGBoost not as a primary predictive model but as a feature space optimization tool operating on GBDT-derived representations. This methodology comprises a two-stage process: first, GBDT is employed for feature derivation, followed by XGBoost for feature selection. This creates a structured pipeline for developing optimized feature representations that maintain a balance between expressive capability and learning efficiency. During the GBDT phase, complex interactions within the data are captured, while the XGBoost phase distills this information down to its most informative elements.

Fig. 6 illustrates this feature screening process. This methodological choice addresses a fundamental challenge in applying complex

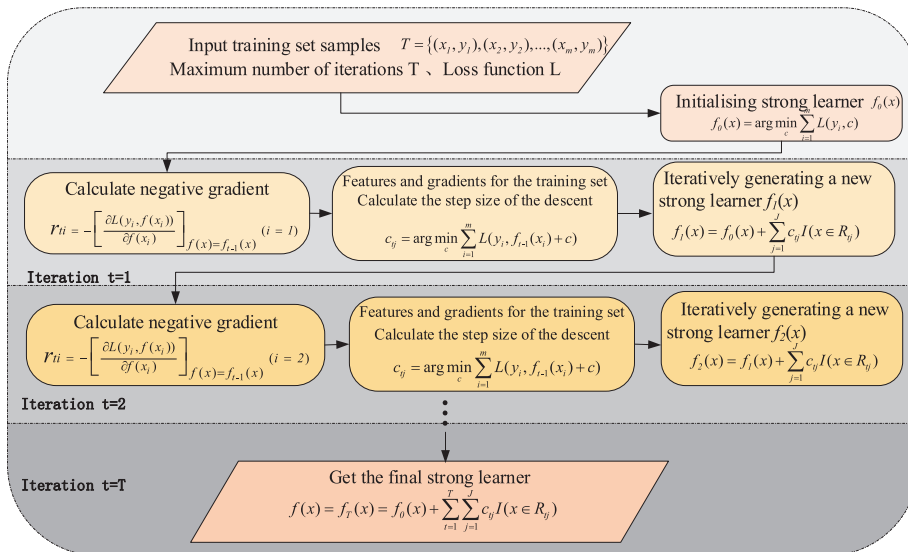


Fig. 4. GBDT algorithm flow.

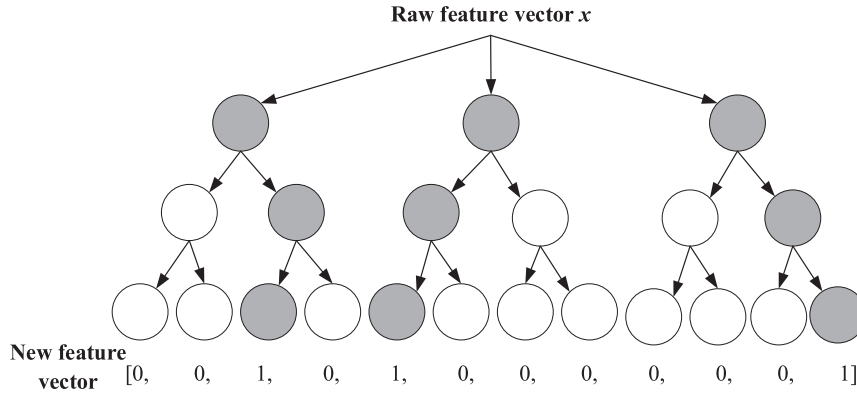


Fig. 5. GBDT feature derivation process.

feature engineering techniques to real-world forecasting problems: the need to manage the trade-off between representation richness and model complexity. By employing XGBoost's embedded feature selection on GBDT-derived features, a feature space is created that preserves the interaction information crucial for accurate prediction while eliminating redundancy and noise that could hinder learning.

3.4. Improved multivariate adaptive multi-grained cascade forest model

This subsection first preprocesses the features affecting the container reloading time based on GBDT and XGBoost algorithms. Subsequently, a multi-granularity scanning structure is used to explore the relationship among multiple variables affecting container reloading time. Finally, the cascade forest structure is improved to solve the issue of subjective selection of forest learners.

The improvement of the cascade forest involves an adaptive learner selection framework grounded in optimization theory. First, the identification of subsequences that maximize inter-subsequence dissimilarity (Algorithm 2) is motivated by ensemble diversity theory, which is known to improve generalization error. Subsequently, assigning the optimal forest learner type to each subsequence is formulated as a canonical assignment problem: given n subsequences and m candidate learner types, find the bijection $\phi : [n] \rightarrow [m]$ that minimizes the total prediction error $\sum_{i=1}^n E_{i, \phi(i)}$, where E is the error matrix. The Hungarian algorithm provides an optimal solution to this problem. Finally, determining the optimal number of each learner type via Bayesian optimization (Algorithm 3) frames hyperparameter search as maximizing the expected improvement $EI(x) = E[\max(f(x) - f(x^+), 0)]$ over a probabilistic surrogate model of the objective function. This data-driven, theoretically-guided configuration tailors the ensemble structure to the specific statistical characteristics of the input data.

More importantly, we specifically develop an improved Multivariate Adaptive Multi-Grained Cascade Forest Model (IM_MAGCFM)

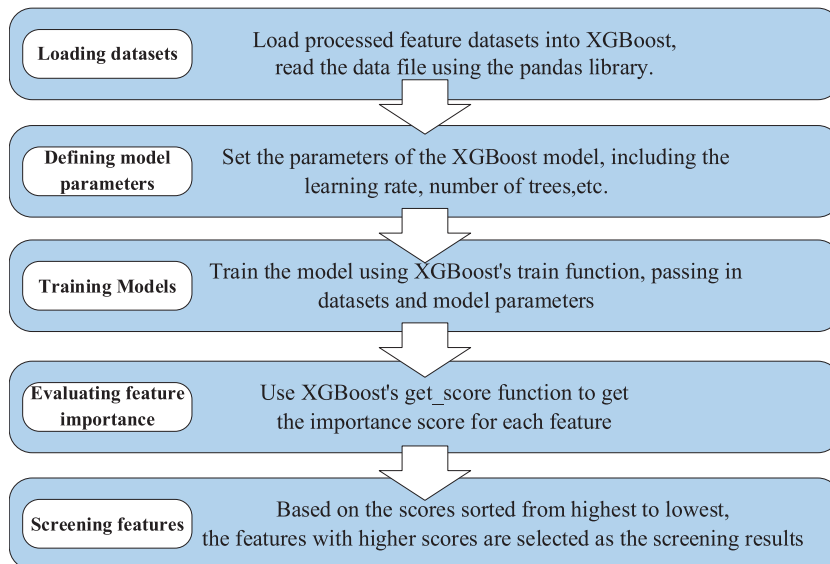


Fig. 6. Steps of feature screening in XGBoost.

for the complex multivariate context of railway ports. The model is capable of extracting more abstract and high-level features that affect the container reloading time, while also uncovering deeper and more hidden relationships within the feature time series. This approach mitigates the challenges associated with predicting container reloading times in complex railway ports, thereby achieving higher accuracy in predictions. Fig. 7 illustrates the flowchart of the IM_MAGCFM-based model for predicting container reloading times at railway ports.

As illustrated in Fig. 7, the IM_MAGCFM model is structured into six main stages.

Step 1: Data preprocessing and feature engineering

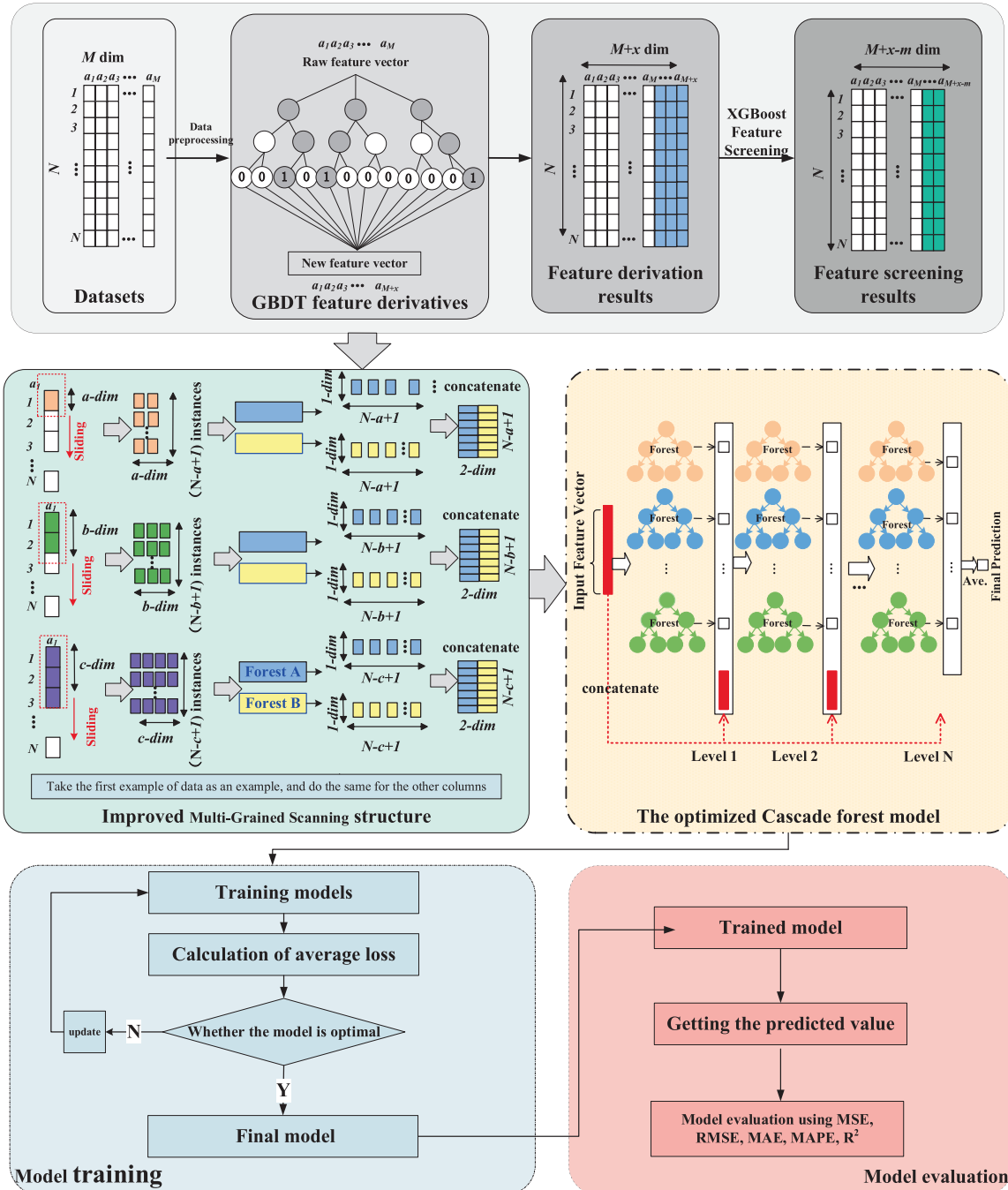


Fig. 7. IM_MAGCFM framework.

In this initial step, the dataset comprising N sets of M -dimension container reloading time feature data undergoes preprocessing. Specifically, continuous variables are addressed using linear interpolation and Lagrange interpolation to manage missing values and outliers. For categorical variables, missing value are filled, and dummy variable coding is applied. Meanwhile, standardization methods are used to harmonize the scale of variable data for subsequent model processing. Then, feature derivation is conducted on the preprocessed data using the GBDT feature derivator, resulting in x derived features. This transformation increases the dimensionality of the container reloading time feature variable from N sets of M -dimensions to N sets of $M + x$ -dimensions. Finally, to eliminate redundant features, the container reloading time variables are ranked using the gain-based XGBoost feature importance assessment method. Based on these importance assessment results, m redundant variables that do not contribute to model performance improvement are eliminated. Consequently, the filtered container reloading time data will consist of N groups of $M + x - m$ -dimensions.

Step 2: Improved multi-grained scanning

In this step, a multi-scale sliding window approach is employed to process the features of container reloading time data, aiming to uncover more hidden feature information. In contrast to traditional multi-grained scanning methods, this paper does not scan the features. Instead, it applies sliding window processing to the samples to uncover hidden data within time series and sequences. Here, we illustrate the specific operation using the first column of data as an example. Assuming a sliding window with a granularity of a is selected for the first column, the sliding window processing yields $N - a + 1$ a -dimensional data points, where a -dimension corresponds to the feature data over every a days. Subsequently, these data points are input into the both random forest and completely random forest, resulting in $N - a + 1$ 1-dimensional outputs. These outputs are then concatenated to form the feature data after multi-grained scanning. Similarly, by setting the granularity to b or c , same operation is performed on the data, and the resulting datasets from multiple granularities are concatenated for input into the cascade forest.

Step 3: Improved cascade forest

This step primarily seeks to mitigate subjectivity in forest learner selection within the cascade forest framework. Initially, **Algorithm 2** identifies subsequences that exhibit maximal differences to enhance forest diversity and determine the optimal types of forest learners. The input sequence data is then segmented into subsequences based on the principle of minimal similarity. These subsequences are predicted independently using candidate forests, resulting in a prediction error matrix that facilitates the identification of the optimal forest learner types. Technically, identifying the optimal forest learner type can be reformulated as an assignment problem, solvable via the Hungarian algorithm. Finally, the optimal number of each type of forest learner is determined using a Bayesian optimization algorithm (**Algorithm 3**).

Table 2

Pseudocode for IM_MAGCFM algorithm.

Algorithm 1:Pseudo-code for IM_MAGCFM algorithm
Input: dataset ($M \text{ dim} * N$) Output: MSE, RMSE, MAE, MAPE, R^2 1: Step1:# Preprocessing categorical and continuous variables 2: for feature $a_1, \dots, a_m$ in categorical_features: 3: apply one-hot encoding to the feature 4: for feature $a_{m+1}, \dots, a_M$ in continuous_features: 5: apply feature scaling to the feature 6: combine processed categorical and continuous_features to form preprocessed_features 7: print preprocessed_features $a_1, \dots, a_M$ 8: Step2:# Feature Derivation with GBDT 9: train a GBDT model on preprocessed_features $a_1, \dots, a_M$ and target variable y 10: merge preprocessed_features $a_1, \dots, a_M$ and new_features $a_{M+1}, \dots, a_{M+x}$ to form combined_features $a_1, \dots, a_{M+x}$ 11: print combined_features $a_1, \dots, a_{M+x}$ 12: Step3:# Feature importance ranking using XGBoost 13: train an XGBoost model on combined_features $a_1, \dots, a_{M+x}$ and target variable y 14: calculate feature importances for all combined_features $a_1, \dots, a_{M+x}$ 15: rank features based on XGBoost importances 16: select top features $a_1, \dots, a_{M+x-m}$ with strong correlation with target_variable y 17: print the top features $a_1, \dots, a_{M+x-m}$ 18: Step4:# Scanning pre-processed features using multi-grained scanning 19: define multi-grained scanning function with three granularity level (a, b, c) 20: perform multi-grained scanning on M column data using the granularity level (a, b, c) 21: concatenate processed data of different granularity 22: print processed_data 23: Step5:# Inputting the results after multi-grained scanning into the improved cascade forest algorithm 24: Input processed_data into the improved cascade forest algorithm 25: train the improved cascade forest algorithm and make final prediction 26: print final prediction and MSE, RMSE, MAE, MAPE, R^2

Step 4: Data partitioning

The purpose of this step is to evaluate the generalization ability of the model on unseen data. In this step, the training and test sets are allocated according to a predetermined ratio. The training set is utilized for model training, while the test set is employed to assess the model's performance.

Step 5: Model training and testing

In this step, the model is trained to identify the best parameters, resulting in the final multivariate container reloading time prediction model. We subsequently test this model against the test set, comparing predictions to actual values to validate its effectiveness.

Step 6: Model evaluation

The final step involves evaluating the model's performance using various metrics, including Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and R-squared as well as assessing the model's performance improvement rate.

3.5. Model algorithm

To enhance understanding of the mechanics behind the constructed model, the pseudocode for the IM_MAGCFM model is presented in Table 2.

In addition, Tables 3 and 4 describe the pseudocodes for sequence splitting and Bayesian optimization algorithm, respectively.

4. Experimental results

In this section, real container reloading time and influencing factor data from Alashankou are used to validate the IM_MAGCFM model proposed in this paper. First, Section 4.1 provides a brief introduction to the algorithmic data. Next, the data processing as well as the feature engineering process is described in Section 4.2. In Section 4.3, the internal parameters of the model are configured. Finally, Section 4.4 selectes and compares several mainstream models, along with improved models, to verify the prediction performance of the IM_MAGCFM model.

4.1. Datasets

To evaluate the predictive performance of the IM_MAGCFM model, this study uses actual container reloading time data from the Alashankou Railway Port. As a major China–Europe rail hub and a key node on the New Eurasia Continental Bridge, Alashankou is highly representative for researching border port operations and supply chain reliability.

The data come from a technical project by the State Railway Administration. Container reloading time is defined as the duration from train arrival until departure after reloading. Factors influencing this time are:

- (1) **Route Congestion:** Measured by queue length; affects turnaround time and reloading efficiency.
- (2) **Heavy Container Rate:** Proportion of loaded containers; higher ratio increases reloading time.
- (3) **Train Full Axle Rate:** Share of fully-loaded axles; indicates maximum capacity utilization.
- (4) **Arrival Station:** Final stop of the train; affects handling time due to local demand.
- (5) **Holiday:** Holidays increase cargo flow and may prolong operations.
- (6) **Weather:** Inclement weather reduces operational safety and efficiency.
- (7) **Extreme Temperatures:** Temperatures beyond 0°C or above 37°C impact equipment and personnel.
- (8) **Unexpected Events:** Includes manifest errors, accidents, or equipment failures causing delays.

These eight variables are used as inputs to predict container reloading time. For modeling, variables are classified as categorical or continuous. Sample reloading time data are provided in Table 5.

Table 3

Pseudo-code for sequence splitting algorithm.

Algorithm 2: Sequence Splitting
Input: A time series dataset X, Δ_1, Δ_2 Output: Number of optimal subsequence divisions n , optimal subsequence set Z_i
1: For $n = \Delta_1$ to Δ_2 do
2: Get subsequences $t_1 \sim t_n$ with δ_i, τ_i
3: the values of the objective function f and the optimal subsequence Z_i are obtained by solving the model M1
4: End for $n = \Delta_1$ to Δ_2
5: Return The value of f and its corresponding number n and the set of optimal subsequences Z_i are recorded during the iterative process.

Table 4

Pseudo-code for Bayesian optimization algorithm.

Algorithm 3: An algorithm for selecting the number of forest learners based on Bayesian optimization	
Input:	
$y=f(x)$ // objective function	
$n_t \in \mathbb{N}$ // number of iterations	
X // search space	
Output: h, MSE	
1: Initial Sampling $D = \{(x^{(1)}, y^{(1)}), \dots, (x^{(k)}, y^{(k)})\}$	
2: Construct probabilistic surrogate model	
3: For $t \leftarrow 1, \dots, n_t$ do	
4: Find x^* that optimizes the acquisition function in X	
5: $D \leftarrow D \cup \{(x^*, f(x^*))\}$	
6: Update probabilistic surrogate model	
7: End for $t \leftarrow 1, \dots, n_t$	
8: Return the number of forest learners corresponding to the smallest objective function	

4.2. Data preprocessing

This section preprocesses the raw container reloading time data to prepare it for model input. The process involves two main steps: handling missing values using linear interpolation and correcting outliers with Lagrangian interpolation. After preprocessing, 424 valid samples remain.

To ensure feature comparability, continuous variables (e.g., heavy container rate and full axle rate) are standardized to zero mean and unit variance, while categorical variables (such as arrival station, holiday, weather, and unexpected events) are one-hot encoded. This scaling improves model training effectiveness. The processed data is shown in Table 6, where a1–a8 correspond to the input features (holiday, arrival station, route congestion, weather, unexpected events, extreme temperature, heavy container rate, and full axle rate), and y represents container reloading time.

To ensure a correlation between the input model's feature variables and the target variable of container reloading time, while also avoiding excessive correlation among the features that could lead to ineffective training, this paper employs correlation analysis to evaluate the associations between the features. The classification of correlation levels among the variables is provided in Table 7.

In order to ensure that there is a correlation between the feature variables of the input model and the target variable of cargo reloading time, and to avoid too much correlation between the features leading to ineffective training, this paper adopts the correlation analysis method to evaluate the association between the features, and the correlation level classification between the variables is given in Table 7.

As can be seen from Table 7, the correlation coefficient ranges from $[-1, 1]$. A correlation coefficient greater than 0 indicates a positive correlation between the two variables, meaning that the value of one variable increases as the value of the other variable increases, and vice versa. The closer the correlation coefficient is to -1 , the stronger the correlation between the two. Through the correlation calculation, the heat map between the characteristic variables of container reloading time and the target variables is obtained, as shown in Fig. 8. In the heat map, colors transition from light to dark, representing a shift from weak to strong correlation.

According to Fig. 8, among the eight input variables, the correlation coefficients between the variables and the target variables are greater than 0, i.e., they are positively correlated with the length of the changeover operation. It means that the larger the values of these characteristics, the longer the time required for the changeover operation. Among them, a₃ line congestion condition, a₈ full axle rate and a₅ unexpected events have the most significant effect on the target variables, while a₄ weather and a₆ extreme temperature show a moderate correlation with the duration of the cargo reloading. In addition to this, the input variables have low correlation with each other, avoiding the problem of duplicated input features.

Table 5

Sample data for time spent in partial container reloading.

ID	Holiday	Arrival station	Route congestion	Weather	Unexpected Events	Extreme Temperatures	Heavy container rate	Train full axle rate	Time
1	N	Tilburg	unclogged	sunny	N	Y	100.00%	70.61%	49
2	Y	Łódź	unclogged	cloudy	Y	Y	97.56%	85.00%	55
3	N	Łódź	unclogged	cloudy	N	N	100.00%	65.61%	39
4	Y	Tilburg	unclogged	Sunny	Y	N	68.29%	74.29%	54
5	N	Nuremberg	General congestion	cloudy	N	Y	92.68%	83.68%	57
...	...	...	...	...	...	...	...	...	...
423	N	Łódź	unclogged	Light snow	Y	Y	97.56%	85.00%	54
424	N	Nuremberg	unclogged	Light rain	N	Y	97.56%	85.00%	53

Table 6

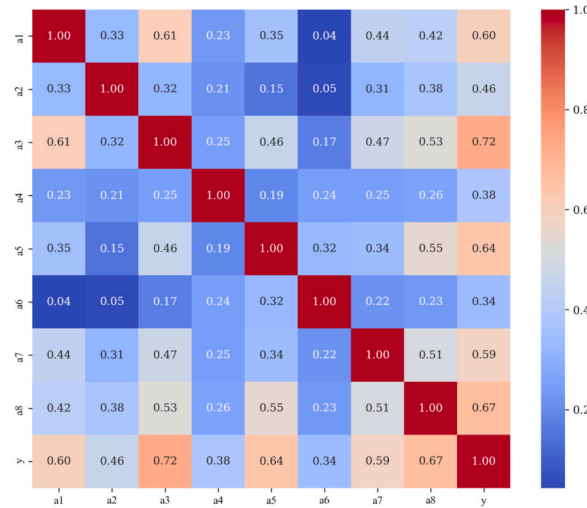
Partial post-processing sample data.

ID	a_1	a_2	a_3	a_4	a_5	a_6	a_7	a_8	y
1	0,1	0,1,0	1,0,0,0	1,0,0,0,0	0,1	1,0	1.01	1.21	1.11
2	1,0	1,0,0	1,0,0,0	0,1,0,0,0	1,0	1,0	1.21	1.21	1.11
3	1,0	1,0,0	1,0,0,0	0,0,1,0,0	0,1	0,1	1.00	-0.09	-0.09
4	0,1	0,1,0	1,0,0,0	1,0,0,0,0	1,0	0,1	-1.09	0.49	0.99
5	0,1	0,0,1	0,1,0,0	0,0,1,0,0	0,1	1,0	0.52	1.12	1.35
...	...	...	...	...	...	...	...	...	...
423	0,1	1,0,0	1,0,0,0	0,0,0,0,1	1,0	1,0	0.85	1.21	0.99
424	0,1	0,0,1	1,0,0,0	0,0,0,0,1	0,1	1,0	0.85	1.21	0.87

Table 7

Correlation coefficient scale.

Relevant propertie	Very strong	Strong negative	Moderate negative	Weak negative	No correlation
Range of negative correlation	[-1, -0.7]	[-0.7, -0.4]	[-0.4, -0.3]	[-0.3, -0.2]	[-0.2, 0]
Range of positive correlation	[0.7, 1]	[0.4, 0.7]	[0.3, 0.4]	[0.2, 0.3]	[0, 0.2]

**Fig. 8.** Heat map of correlation between feature variables.

4.3. Feature engineering

To effectively extract more abstract features from the sample data and enhance the modeling capability of the model, this paper uses the GBDT feature derivator for feature derivation on the original data. Following this process, the feature dimension is increased by 100, enabling the model to better capture the complex relationships within the data and thereby improve its predictive performance and generalization ability. Meanwhile, to eliminate redundant features that have minimal impact on the model effect, we employ the XGBoost feature importance assessment method to rank the importance of the newly derived features, of which the top 30 features are shown in Fig. 9.

A higher feature importance score indicates that a feature contributes more significantly to the predictive model. Therefore, after ranking the derived features, those that contributed less to the model were eliminated. Fig. 9 displays only the top 30 derived features based on their importance rankings. From the Fig. 9, it is evident that the derived feature a_{11} has the highest importance, with a score of 173. The scores for a_{19} , a_{47} , and a_{23} are all above 100, indicating a substantial contribution to the model. The remaining derived features have importance scores value of 30 or higher, suggesting a certain level impact on the model. The derived features ranked in the top 30 for importance were then combined with the original revised features to create the preprocessed model features.

4.4. Parameters setting

4.4.1. Optimal number of forest learner types

As shown in Fig. 10, the lower limit for forest learner type Δ_1 is set to 2 to ensure the diversity among the forest learners. Additionally, considering that the width of each level in the deep forest should not be excessively large, the value of Δ_2 is set to 5. The

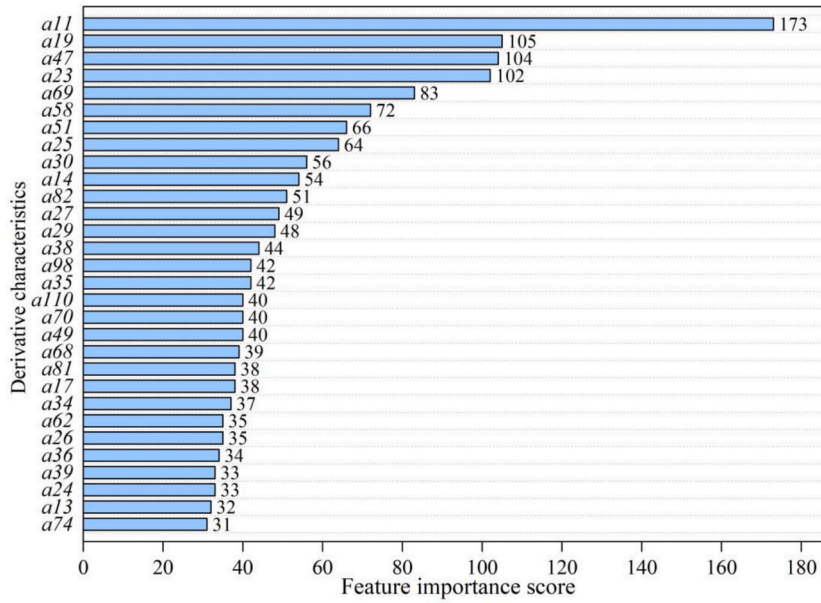


Fig. 9. Results of the assessment of the importance of some features.

optimization problem of sequence division is solved by Algorithm 2, which records the value of f during the iteration process, along with the corresponding number of subsequences n and the optimal set of subsequences Z_i , as illustrated in Fig. 10.

Fig. 10 demonstrates the iterative process of optimizing the subsequence division of the time series data for the container reloading at the Alashankou railway port. It is evident that the value of f reaches its maximum and the average similarity between the divided subsequences is minimized when $n = 4$. Therefore, the optimal number of forest learner types is established at 4, based on the real container reloading time data at Alashankou railway port. In addition, Fig. 11 showcases the subsequence that emerge from the time series data when we set n to 4.

4.4.2. Optimal type of forest learner

In order to obtain the prediction error matrix, forest learners are used to process the dressing job time subseries depicted in Fig. 11 separately. Considering diversity and broad applicability, five common forest models based on decision tree integration are selected as candidate forest learners: Random Forest (RF), Completely Random Forest (CRF), Gradient Boosting Decision Tree (GBDT), eXtreme Gradient Boosting (XGB), and Light Gradient Boosting Machine (LGBM). Table 8 shows the prediction errors for container reloading time subsequence at the Alashankou railway port across these five candidate forest learners.

By applying the Hungarian method solution model to the forest selection problem outlined in Table 8, the corresponding candidate forest learner for each subsequence can be obtained. Specifically, subsequence t_1 achieves the smallest prediction error under the XGB model, with a result of 12.746; subsequence t_2 has the smallest prediction error under the CRF model, with a result of 12.859;

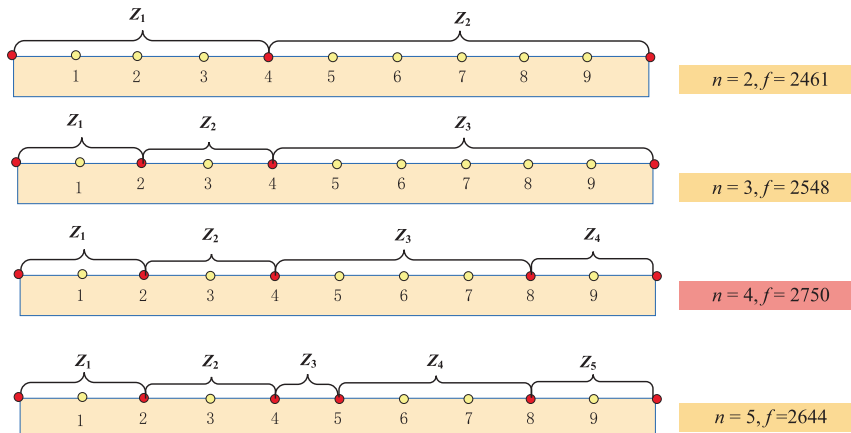


Fig. 10. Record of the iterative process of optimal subsequence division.

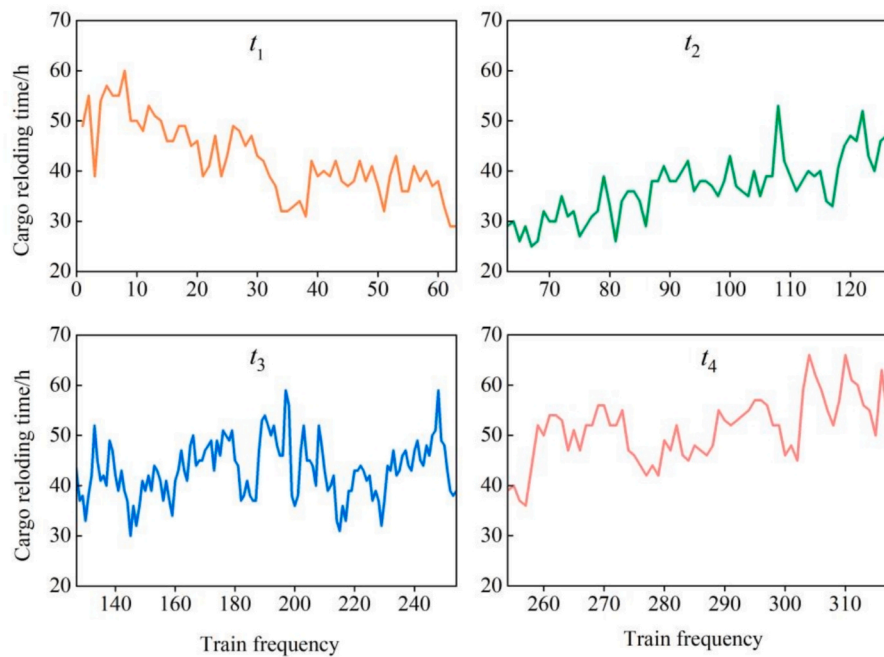


Fig. 11. Optimal subseries set of Alashankou railroad port container reloading time data.

subsequence t_3 performs best under the RF model, with a result of 9.029; and subsequence t_4 shows the smallest prediction error under the LGBM model, with a result of 12.482. Consequently, the final four forest learner types selected for the Alashankou railway port container reloading time series are XGB, CRF, RF and LGBM.

4.4.3. Optimal number of each type of forest learner

Based on the optimal types of forest learners obtained above, Algorithm 3, the Bayesian optimization method, is used to determine the optimal number of forest learners for the selected four types. The search space for each type of forest learner ranges from [1, 4]. By iterating through Algorithm 3, the prediction error variation of the model for the four types of forest learners—RF, CRF, XGB and LGBM—at different values is obtained, as illustrated in Fig. 12.

As shown in Fig. 12, all three forest learners—RF, CRF and XGB—converge at a value of 1, indicating that the prediction error of the model is minimal at this point. Additionally, the MSE value for LGBM converges within the interval of two neighboring points, with a relatively low prediction error value when the number of LGBM is between 2 and 3. Meanwhile, it is evident from Fig. 12 that the model's prediction error value reaches its minimum when the number of RF, CRF and XGB is equal to 1, while the number of LGBM is equal to 3.

Through a series of parameter experiments, the optimal values for key hyperparameters have been determined. Notably, the granularity design set [1, 7, 14] is not arbitrary; it is directly informed by the multi-granularity temporal patterns of railway port operations discussed in Section 1.2. These values represent micro (single train run), meso (weekly cycle), and macro (bi-weekly phase) granularities, ensuring that the multi-grained scanning mechanism aligns with the operational cadence. To further enhance the model's prediction performance, additional experiments were conducted on parameters such as the number of decision trees within each forest. The final, optimized parameter set for the IM_MAGCFM model is presented in Table 9.

4.5. Results and discussions

In this subsection, multiple types of prediction models are used to forecast container reloading time at Alashankou for evaluating the performance of the proposed IM_MAGCFM model. The comparison models include three types: univariate prediction models,

Table 8

Prediction errors for the time subseries of container reloading on the five candidate forest learners.

MarqueeForestSubsequence	RF	CRF	GBDT	XGB	LGBM
t_1	12.953	13.286	13.258	12.746	13.549
t_2	13.128	12.859	13.028	13.102	13.426
t_3	9.029	9.58	9.213	9.547	9.46
t_4	12.816	13.452	14.126	12.798	12.482

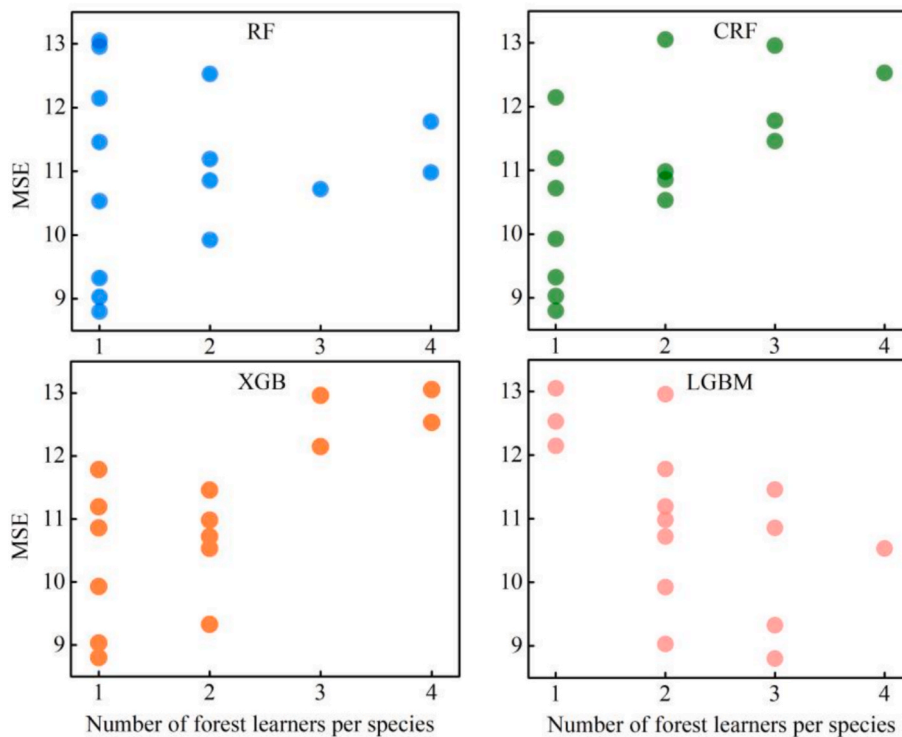


Fig. 12. Optimization process for the optimal number of each forest learner.

Table 9
Optimal hyperparameters in the IM_MAGCFM model.

Name	Hyperparameter
Scanned granularity a, b, c	1,7,14
Type of forest learner	RF, CRF, XGB, LGBM
Number of forest learner per layer	6
Combination of forest learner	RF, CRF, XGB, LGBM, LGBM, LGBM
Number of decision tree	500

mainstream multivariate models, and multivariate models enhanced by feature engineering. Table 10 presents a selection of prominent multivariate models, including Convolutional Neural Network (CNN), Long Short-Term Memory (LSTM), Graph Convolution Network (GCN), Informer-LSTM, XGB, and LGBM.

Table 11 presents improved multivariate cascade forest variants: the base Multi-Grained Cascade Forest (MGCF), MGCF with Feature Derivation (MGCF_FD), MGCF with Feature Screening (MGCF_FS), the Multivariate Adaptive Multi-Grained Cascade Forest with enhanced scanning (IM_MGCF), and the proposed IM_MAGCFM model.

4.5.1. Evaluation of metrics performance

The IM_MAGCFM model was used to predict the multivariate container reloading time at the Alashankou railway port station. Concurrently, the same training and test data were utilized to predict the comparison models listed in Tables 10 and 11. Table 12 outlines the evaluation metrics for the comparison prediction models as well as the proposed IM_MAGCFM model, which include MSE,

Table 10
Basic characteristics of current mainstream multivariate prediction models.

Comparison model	Characteristic
CNN	Capable of extracting local temporal patterns through convolutional filters, capturing short-term dependencies in time series data.
LSTM	Designed to model sequential data and capture long-term dependencies via memory cells and gating mechanisms.
GCN	Capable of handling complex and multivariate timing data, capturing dependencies between nodes.
Informer-LSTM	Can adaptively model data at different time scales, capturing long-term dependencies and spatial correlations.
XGB	Automatically selects the best features and hyperparameters and can handle complex nonlinear relationship.
LGBM	Fast to train and predict and can handle large-scale data.

Table 11

Basic characteristics of improved multivariate multi-granularity cascade forest correlation models.

Contrast model	Multivariate	Feature derivation	Feature screening	Improved multi-grained scanning	Adaptive base learner
MGCF	✓	×	×	×	×
MGCF_FD	✓	✓	×	×	×
MGCF_FS	✓	✓	✓	×	×
IM_MGCF	✓	✓	✓	✓	×
IM_MAGCFM	✓	✓	✓	✓	✓

RMSE, MAE, and MAPE. In addition, the degree of improvement in the prediction performance of the IM_MAGCFM model compared to the other comparison models is provided for each metric.

Based on the data presented in Table 12, the IM_MAGCFM model proposed exhibits excellent prediction performance in the field of multivariate prediction, outperforming the current mainstream models. The quantitative enhancements in error metrics directly enhance operational predictability. For example, the significant reduction in Mean Squared Error (MSE) indicates a lower average forecast deviation, leading to more reliable scheduling windows. This increased forecasting accuracy minimizes the need for excessive safety buffers in operational plans, potentially boosting asset utilization and terminal throughput. These significant performance improvements are mainly attributed to the model's effectiveness during the feature processing stage. Notably, IM_MAGCFM achieves a 78.07% reduction in MSE compared to CNN and a 72.16% reduction compared to LSTM, showcasing its superior ability to model complex data from multiple time series. Compared to other models, IM_MAGCFM is more adept at capturing the hidden feature information within the original data and processing feature details with greater depth. In addition, while the XGB and LGBM models are more sensitive to outliers, IM_MAGCFM demonstrates greater robustness in this regard. Collectively, these advantages prove the necessity and effectiveness of the proposed IM_MAGCFM model. To further emphasize the importance of the feature processing stage, Fig. 13 presents the results of the prediction evaluation metrics for each model based on the improved multivariate multi-grained cascade forest.

As shown in Table 12 and Fig. 13, the prediction error of the improved model based on the multivariate multi-grained cascade forest (MGCF) consistently decreases through ongoing enhancements. Following feature screening (MGCF_FS) and feature derivative (MGCF_FD), the model's prediction error is further minimized. This reduction is attributed to feature derivation, which uncovers hidden relationships among variables, while feature screening enhances the model's efficiency and generalization capability. After refining the multi-grained scanning structure, the error metric of the IM_MGCF model decreases significantly. This improvement results from the transition from scanning features to scanning samples, allowing the model to more effectively extract temporal relationships. Ultimately, the IM_MAGCFM model incorporates an adaptive tuning forest learner function into the IM_MGCF, enabling it to identify the optimal combination of base learners based on the input data. Its MSE, RMSE, MAE, and MAPE values are 3.43, 1.85, 1.51, and 1.12, respectively, demonstrating superior prediction results compared to other models.

4.5.2. Performance of fitting

Table 13 presents the values of the coefficient of determination (R^2) for various types of prediction models, indicating that the closer the value R^2 is to 1, the better the model fit.

Additionally, Fig. 14 compare the predicted values obtained by the IM_MAGCFM model and each comparison model against the actual values for container reloading time at the Alashankou railway port. The first 75% of the dataset on the horizontal coordinates represents the model's training set, while the last 25% corresponds to the test set. To clearly illustrate the fitting performance of the different models on the test set, a more detailed section of it is presented in the lower part of each figure.

Table 13 and Fig. 14 provide the following insights:

- (1) From Table 13 and Fig. 14(a) and (b), the R^2 of GCN and Informer-LSTM are 0.51 and 0.60, respectively, indicating relatively poor fitting effects. In contrast, the IM_MAGCFM model achieves an R^2 value of 0.86, demonstrating superior fitting performance compared to these multivariate deep learning models.
- (2) As shown in Fig. 14 (c) and (d), the XGB and LGBM models exhibit improved fitting effects compared to deep learning models; however, they primarily predict general trends, with most predicted values falling below the actual values.
- (3) Following feature derivation (MGCF_FD) and feature filtering (MGCF_FS), the fitting performance of the MGCF model gradually improves. Additionally, enhancements to the multi-granularity scanning structure increases the R^2 values from 0.73 and 0.76 to 0.82, with the IM_MGCF model demonstrating closer fitting performance at critical nodes. The IM_MAGCFM model proposed in this paper achieves an R^2 value of 0.86, outperforming the other compared models and showcasing its superiority in fitting performance at critical nodes.

4.6. Generalization experiment

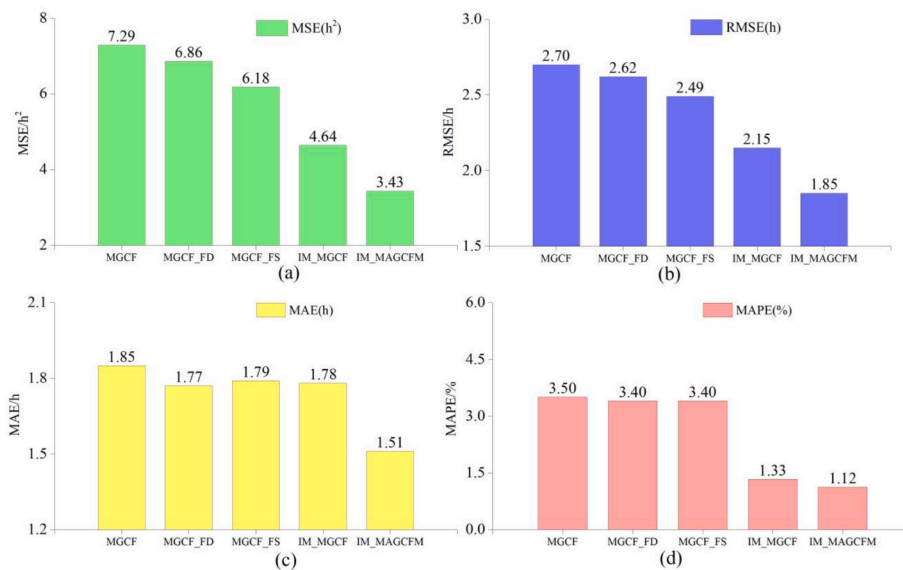
To evaluate the generalization capability of the proposed IM_MAGCFM model, this study conducted experiments using a real-world railway border crossing dataset. The data originates from Małaszewicze Port, another critical border station for CRE trains, and includes container transshipment times and their influencing factors.

The data processing workflow aligns with Section 4.2. Experiments compared IM_MAGCFM against multiple state-of-the-art

Table 12

Evaluation metrics for different models predicting multivariate container reloading time.

Model	MSE /h ²	IM_MSE /%	RMSE /h	IM_RMSE /%	MAE /h	IM_MAE /%	MAPE /%	IM_MAPE /%
CNN	15.64	78.07	3.96	53.28	3.29	54.10	6.25	82.08
LSTM	12.32	72.16	3.51	47.29	3.01	49.83	5.65	80.18
GCN	12.31	72.14	3.51	47.29	2.67	43.45	5.10	78.04
Informer-LSTM	10.01	65.73	3.16	41.46	2.37	36.29	4.49	75.06
XGB	8.69	60.53	2.95	37.29	2.21	31.67	4.20	73.33
LGBM	7.84	56.25	2.80	33.93	2.18	30.73	4.00	72.00
MGCF	7.29	52.94	2.70	31.48	1.85	18.46	3.50	68.00
MGCF_FD	6.86	49.96	2.62	29.26	1.77	14.63	3.40	67.06
MGCF_FS	6.18	44.52	2.49	25.53	1.79	15.35	3.40	67.06
IM_MGCF	4.64	26.01	2.15	13.98	1.78	14.92	1.33	15.79
IM_MAGCFM	3.43		1.85		1.51		1.12	

**Fig. 13.** Prediction errors of various types of models based on improved multivariate multi-grained cascade forests.**Table 13** R^2 values for different prediction models.

Model	GCN	Informer-LSTM	XGB	LGBM	MGCF	MGCF_FD	MGCF_FS	IM_MGCF	IM_MAGCFM
R^2	0.51	0.60	0.66	0.69	0.72	0.73	0.76	0.82	0.86

models, including deep learning models (CNN, LSTM), ensemble learning models (LGBM, GBDT, RF), and advanced forest models (DF, NET-DF, DEF, MGCF). Results are presented in Table 14, where IM_* metrics denote the percentage gap between each model's performance and IM_MAGCFM.

The generalization experiments conducted on the Malaszewicze Port dataset provide compelling evidence that the proposed IM_MAGCFM model demonstrates outstanding cross-scenario adaptability and robustness. As shown in Table 14, under completely independent data conditions, IM_MAGCFM not only maintains comprehensive performance leadership but also achieves an additional 10.63% improvement in MSE compared to the strongest baseline model MGCF. This result highlights the universal advantages of its underlying architecture. The findings clearly indicate that the model's core mechanisms—multi-granularity scanning based on samples and adaptive forest learner configuration—are not merely overfitting optimizations tailored to specific datasets. Instead, they universally capture the complex, dynamic temporal dependencies and feature interaction patterns inherent in diverse port operations.

Experiments demonstrate that IM_MAGCFM effectively transfers and adapts to new operational environments and data distributions, consistently delivering stable performance advantages across diverse scenarios. This confirms that the framework's applicability extends beyond the specific conditions of Alashankou Port and holds potential for broader deployment in railway hubs and multi-source heterogeneous data forecasting tasks. It provides critical justification for its implementation in practical logistics scheduling systems.

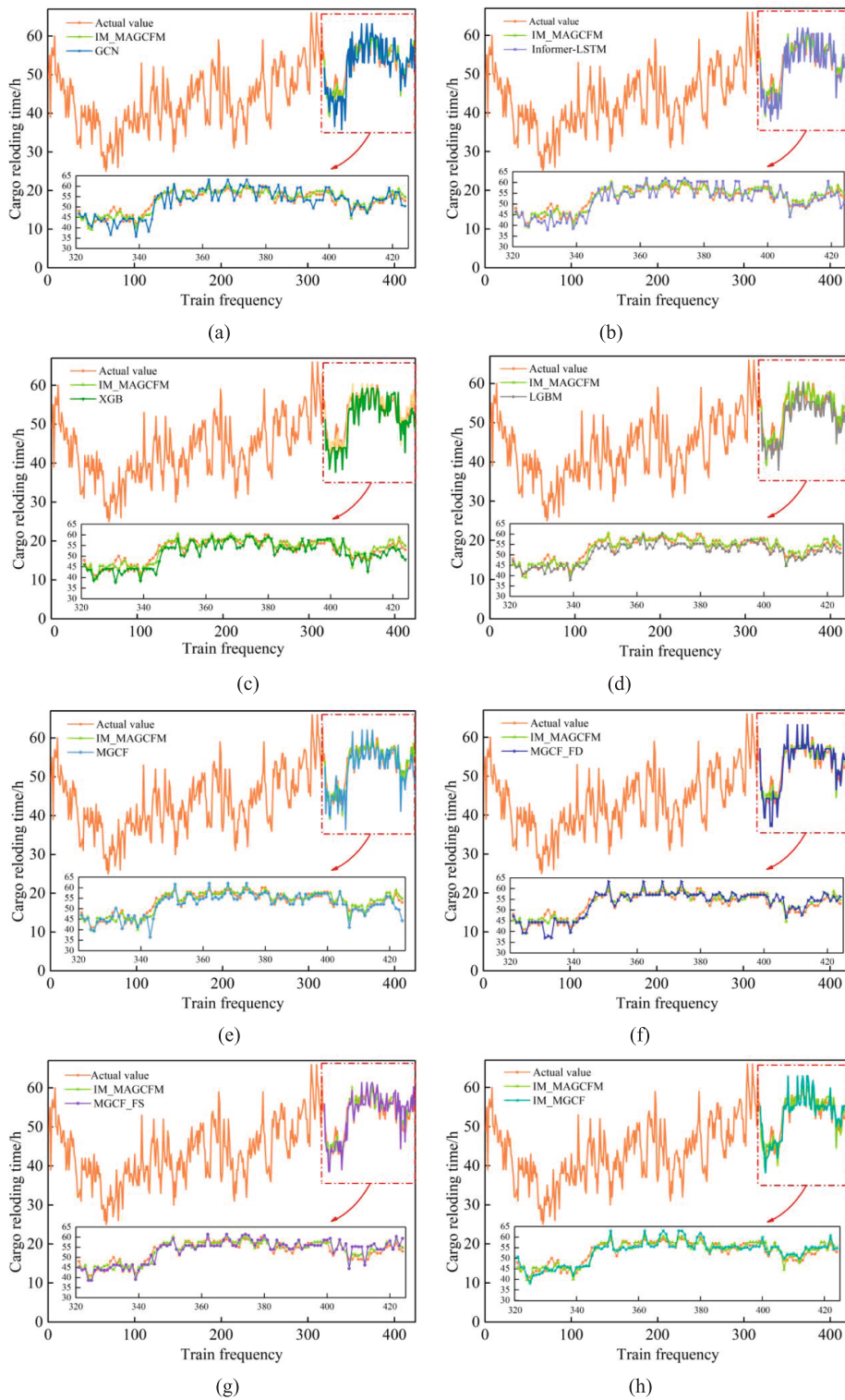


Fig. 14. Plot of true values vs. predicted values from IM_MAGCFM model and other comparative models.

Table 14
Comprehensive evaluation metrics of different models on the Malaszewicze dataset.

Model	MSE /h ²	IM_MSE /%	RMSE /h	IM_RMSE /%	MAE /h	IM_MAE /%	MAPE /%	IM_MAPE /%
CNN	11.94	65.49	3.46	40.75	2.76	43.48	11.25	42.22
LSTM	8.34	50.60	2.89	29.76	2.32	32.76	9.66	32.71
LGBM	7.14	42.02	2.67	22.85	2.15	24.36	8.66	24.94
GBDT	6.84	39.77	2.61	20.69	2.06	19.87	8.48	23.35
RF	5.94	30.64	2.44	14.75	1.93	13.47	7.98	18.55
DF	5.83	29.33	2.41	13.69	1.88	11.70	7.69	15.47
NET-DF	5.62	26.69	2.37	11.81	1.85	9.73	7.64	14.92
DEF	5.40	23.70	2.32	9.48	1.80	6.98	7.48	13.10
MGCF	4.61	10.63	2.15	5.58	1.67	6.59	6.67	2.55
IM_MAGCFM	4.12	—	2.03	—	1.56	—	6.50	—

Note: NET-DF represents Novel Ensemble Tree-Deep Forest; DEF represents Deep Ensemble Forest.

5. Conclusions

The prediction of container reloading time at railway ports constitutes a complex multivariate forecasting challenge influenced by numerous interacting factors. Traditional forecasting approaches exhibit limitations in addressing the inherent complexities of this multi-source operational data structure. This study addresses these limitations through the development of an Improved Multivariate Adaptive Multi-Granularity Cascade Forest Model (IM_MAGCFM), specifically designed for container reloading time prediction in large railway port environments.

Experimental validation using operational data from the Alashankou railway port demonstrates the model's efficacy. The IM_MAGCFM significantly outperforms current mainstream forecasting methods, including deep learning architectures (GCN, Informer-LSTM), ensemble learning models (XGB, LGBM), and enhanced cascade forest variants. Quantitative results show superior performance across all evaluation metrics, with MSE, RMSE, MAE, and MAPE values of 3.43, 1.85, 1.51, and 1.12 respectively, representing substantial improvements over baseline models. The model's coefficient of determination ($R^2 = 0.86$) further confirms its exceptional fitting capability.

Furthermore, generalization experiments conducted on the Malaszewicze border crossing dataset further validate the robustness of IM_MAGCFM. This model not only comprehensively outperforms all baseline models but also demonstrates significant advantages over the state-of-the-art MGCF model across all key metrics. This cross-border validation demonstrates that IM_MAGCFM can effectively handle diverse multi-variable time series structures and operational scenarios, showcasing its robust potential for practical deployment across varied railway border environments.

The findings of this study carry not only methodological significance but also important implications for railway port operational management practices. At the operational level, the improvement in prediction accuracy enables more reliable estimation of reloading operation times, providing precise time windows for dynamic resource scheduling. Port managers can utilize the prediction results to optimize the allocation of cranes, transport vehicles, and human resources, facilitating a transition from reactive response to proactive planning management models that effectively address the uncertainty of operational demands.

At the strategic level, this predictive model provides quantitative foundations for port capacity analysis, bottleneck identification, and long-term infrastructure investment planning. By reducing prediction uncertainty, the model contributes to enhancing the reliability of train schedules—a critical performance metric for railway freight corridors and an important determinant of supply chain resilience. More accurate operation time predictions can strengthen collaboration and coordination among various participants in the logistics chain, ultimately improving the overall service quality and market competitiveness of the CRE.

Despite significant progress, this study still presents several potential weaknesses. First, while the model effectively handles available operational factors, incorporating additional influential elements such as detailed seasonal patterns and personnel skill metrics could further enhance prediction accuracy. Second, extending the IM_MAGCFM framework to related forecasting tasks in railway logistics—such as train running time prediction at intermodal hubs or station dispatching operations—represents a promising direction for broader applicability and impact. These considerations establish a clear pathway for advancing forecasting methodologies in transportation logistics and operational research.

In summary, through methodological innovation and empirical validation, this study not only advances the technical frontier of complex time series forecasting but, more importantly, provides practical and feasible solutions for improving the operational efficiency of railway ports and enhancing the reliability of international logistics corridors. The research results demonstrate that the forecasting framework based on multi-scale feature learning and adaptive integration can effectively address the complex dynamic characteristics of port operations, providing robust decision support for the high-quality development of the CRE.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GPT 5.2 in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRediT authorship contribution statement

Jingwei Guo: Writing – review & editing, Writing – original draft, Validation, Methodology, Conceptualization. **Yimin Wang:** Writing – review & editing, Writing – original draft, Validation, Software, Methodology, Conceptualization. **Zhen-Song Chen:** Writing – review & editing, Validation, Methodology, Funding acquisition, Conceptualization. **Francisco Chiclana:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Witold Pedrycz:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data cannot be made publicly available due to confidentiality restrictions..

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